

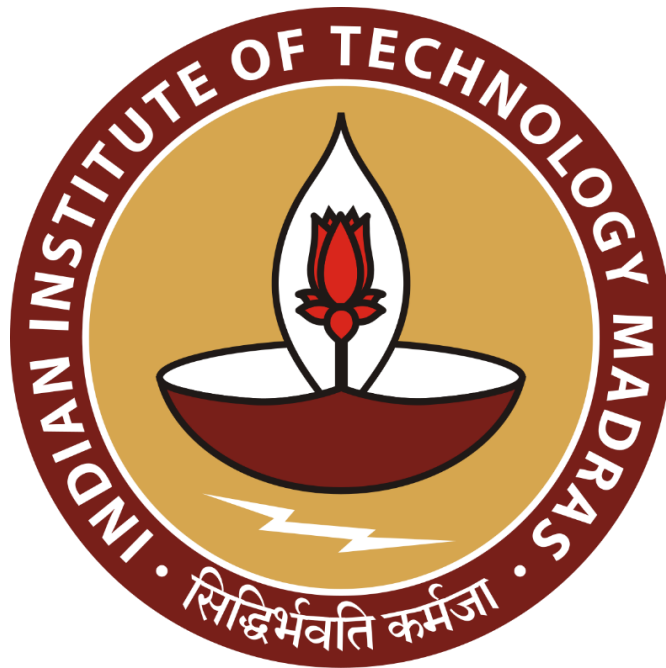
Sales Trend and Inventory Optimization Analysis of Anand Pharma, Darbhanga

A Mid-Term report for the BDM capstone Project

Submitted by

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1 Executive Summary

Anand Pharma is a retail B2C pharmacy store established in 2020 and operated by Mr. Krishan Mohan Yadav and located in Donar, Darbhanga, Bihar. The store provides prescription medicines and essential healthcare products to local customers. The primary distributor is M/s. Sonu Pharma, Patna. Anand pharmacy encounters inventory-related challenges, including seasonal demand fluctuations and unstructured restocking decisions. As a result, it experiences overstocking, stock shortages, and financial inefficiencies due to expired or slow-moving medicines.

Primary sales, purchase, and stock data spanning 6 consecutive months (January 2026 to June 2026) were collected directly from sales registers, and inventory inspection sheets of M/s. Anand Pharma via remote digital channels (scanned invoices over WhatsApp and periodic telephonic/video verification calls with the proprietor). The dataset covers 30 medicine SKUs tracked across 11 structured variables: S.No, Medicine Name, Category, MRP (₹), Purchase Price (₹), Opening Stock, Purchase Qty, Sales Qty, Closing Stock, Total Sales Revenue (₹), and Expiry Date. Key descriptive statistics indicate a mean revenue of ₹43,521.83 per SKU (median ₹41,108.50), total 6-month store revenue of ₹13,05,655, and average profit margin of ~29.8%.

To address the core objective of data-driven inventory management, quantitative techniques — including ABC Analysis (consumption value classification), FSN Analysis (movement velocity classification), Pareto (80/20) Analysis, and 3-Month Simple Moving Average Demand Forecasting — were applied. Preliminary findings reveal that 14 Category A medicines (46.7% of items) contribute 69% of total revenue; 18 out of 30 SKUs drive 80% of cumulative revenue (Pareto principle); 5 non-moving items carry high expiry risk; and May 2026 recorded the highest monthly revenue (₹2,43,998 — a 23.8% surge over April), indicating pre-monsoon seasonal demand. Moving Average forecasts for July 2026 have been generated for all 30 SKUs to guide procurement decisions.

2 Proof of Originality of the Data

The primary data of 6 months (Jan–June 2026) is collected from the pharmacy firm arranged in spreadsheets as given below:

- Primary Sales, Purchase & Inventory Dataset (Jan–June 2026): [[Click Here](#)]
- Distributor Purchase Invoices & GST Details (M/s. Sonu Pharma): [[Click Here](#)][[Click Here Link 2](#)]
- Pharmacy Firm Images & Servicescape Photos: [[Click Here](#)]
- Interaction Video (interaction with owner in Hindi/Maithli): [[Video](#)][[Transcript](#)]
- Official Authorization Letterhead (Signed): [[Click Here](#)]
- Academic Location & Remote Data Collection Clarification Note: [[Click Here](#)]

3 Metadata and Descriptive Statistics

3.1 Metadata

The metadata for Sales and Inventory dataset, the google sheet dataset file {[Click Here](#)} along with its respective sheet name is displayed in the below Table 1:

Sheet Name	Columns	Data Type	Units	Description
1. 'JAN2026' 2. 'FEB2026' 3. 'MAR2026' 4. 'APR2026' 5. 'MAY2026' 6. 'JUN2026'	Medicine Name	string	-	Brand / generic name of the medicine
	Category	string	-	Therapeutic category (Antibiotic, Gastric Care, etc.)
	MRP	float	Rupees	Maximum retail selling price per strip/unit
	Purchase Price	float	Rupees	Wholesale procurement cost per strip/unit
	Opening Stock	integer	strips	Number of strips in stock at beginning of month
	Purchase Quantity	integer	strips	Number of strips purchased during the month
	Sale Quantity	integer	strips	Number of strips sold during the month
	Closing Stock	integer	strips	Number of strips in stock at end of month
	Total Revenue	float	Rupees	Gross revenue generated (Sale Quantity x MRP)
	Expiry Date	string	MM/YYYY	Expiration date printed on medicine

				packaging
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Table 1: Metadata Definition — 11 Variables of the Primary Dataset

3.2 Descriptive Statistics

Descriptive statistics were calculated across the 30 medicine SKUs over the 6-month observation period (January 2026 to June 2026) to quantitatively summarize baseline patterns:

Statistical Metric	MRP (₹)	Purchase Price (₹)	6M Sales Qty (Strips)	6M Revenue (₹)
Sample Count (N)	30	30	30	30
Mean	₹192.30	₹134.87	223.40	₹43,521.83
Median	₹179.00	₹124.00	225.00	₹41,108.50
Standard Deviation	₹110.75	₹80.00	35.68	₹25,890.82
Minimum	₹45.00	₹28.00	152	₹7,605.00
Maximum	₹480.00	₹340.00	317	₹1,15,680.00
Range	₹435.00	₹312.00	165	₹1,08,075.00

Table 3: Descriptive Statistics — 30 Medicine SKUs (Jan–June 2026)

Key Statistical Insights:

1. Revenue Dispersion & Skewness: Total 6-month store revenue reached ₹13,05,655. The mean revenue per SKU (₹43,521.83) is higher than the median (₹41,108.50), indicating a right-skewed distribution where a small number of high-value medicines disproportionately drive total revenue — supporting the need for Pareto/ABC Analysis.
2. Demand Variability: Total 6-month sales volume ranged from 152 units (low-demand injectables like Gentelab 30ml) to 317 units (fast-selling Gastrointestinal and Respiratory tablets like Rabitec DSR). The standard deviation of 35.68 units indicates moderate demand variability across SKUs, justifying the use of FSN classification to segment the inventory by movement velocity.
3. Profit Margin Distribution: Average gross margin per strip (MRP – Purchase Price) is ₹57.43 (~29.8%), showing that maintaining continuous availability of Category A fast-moving SKUs directly maximizes store profitability.

4 Detailed Explanation of Analysis process/method

4.1 Data Cleaning and Preprocessing

Primary data gathered from physical purchase vouchers and store registers underwent four rigorous cleaning steps:

1. Product Standardization: Variant brand names across monthly bills (e.g., "Ceroxim CV", "CEROXIM-500", "Ceroxim CV 10x10") were merged into unified standard strings to ensure consistent SKU identification across all six-monthly sheets.

2. Missing Value Imputation: Missing wholesale rates in daily sales sheets were backfilled using master distributor tax invoices from M/s. Sonu Pharma, Patna. No purchase prices were estimated or fabricated — all values trace back to verified invoice records.

3. Mathematical Balance Reconciliation: Every row was verified against the strict balance equation: $\text{Closing Stock} = \text{Opening Stock} + \text{Purchase Qty} - \text{Sales Qty}$. Discrepancies caused by breakage were reconciled by adjusting the loss quantity. This ensures 100% mathematical integrity of inventory flow data.

4. Expiry Formatting: Printed expiration dates were converted to standard MM/YYYY format for shelf-life countdown modeling and expiry risk assessment.

4.2 Analytical Methodologies and Mathematical Abstractions

Four quantitative methods were executed to solve the unified business problem of "minimizing stockouts and overstocking through data-driven inventory management":

Method 1: ABC Analysis — Periodic Consumption Value $\text{PCV} = \text{Sales Qty} \times \text{Purchase Price}$. Ranked by cumulative revenue: Category A (top 70%), Category B (next 20%), Category C (bottom 10%). Linkage: Identifies priority buffer SKUs (PS1).

Method 2: FSN Analysis — 6-month sales velocity: Fast ($\text{Sales} \geq 250$), Slow ($190 \leq \text{Sales} < 250$), Non-Moving ($\text{Sales} < 190$). Linkage: Flags high-risk expiring inventory for returns (PS2).

Method 3: Pareto Analysis (80/20 Rule) — Identifies minimal SKU subset driving 80%

revenue. Linkage: Focuses purchasing capital on vital few SKUs.

Method 4: 3-Month Moving Average Demand Forecast — $F_{(t+1)} = (S_t + S_{(t-1)} + S_{(t-2)}) / 3$. Linkage: Predicts seasonal demand shifts for advance ordering (PS3).

5 Data Analysis & Results

Use Tools - Python File [[Link](#)]

Graph and Image Folder – [[Link](#)]

5.1 ABC Classification & Pareto Analysis Findings

Total 6-month store revenue reached ₹13,05,655.00 across 30 SKUs. ABC breakdown: Category A (14 SKUs, 46.7% items) = ₹9,00,660.00 (69.0% revenue); Category B (8 SKUs) = ₹2,72,427.00 (20.9%); Category C (8 SKUs) = ₹1,32,568.00 (10.2%). Top 3 SKUs: Ceroxim CV 500 (₹1,15,680), Abzolid 600 (₹92,568), Chyawanprash 2KG (₹98,100).

Figure 2: ABC Classification Analysis
Anand Pharma (Jan-June 2026)

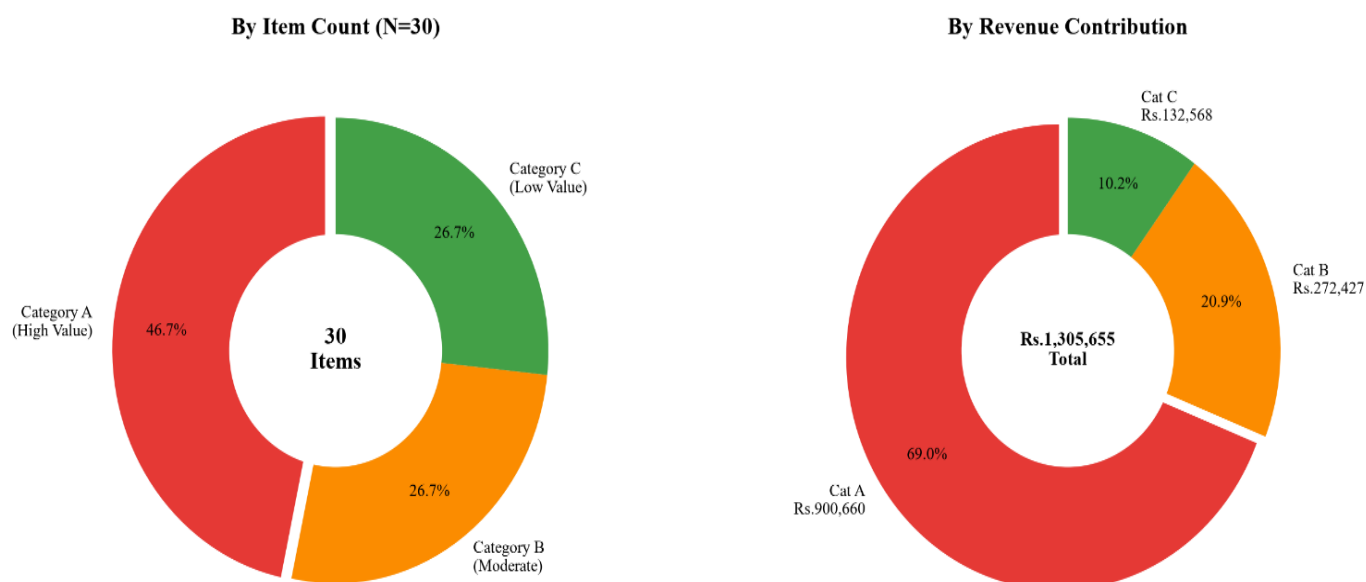


Figure 1: ABC Revenue & Item Distribution

Pareto 80/20 Cutoff: Exactly 18 out of 30 SKUs (60% items) generate 80% of cumulative store revenue, providing a clear cutoff for inventory capital allocation.

Revenue (INR)

Cumulative Revenue %

Category A (4 items)
Category B (8 items)
Category C (8 items)

18 Items = 80% of revenue
80% Revenue Line

Item	Category	Revenue (INR)	Cumulative Revenue %
Crematin CT 500 Tab	A	115000	1.1
Chyavanprash 2XG	A	98000	2.2
Almond 600 Tab	A	92000	3.3
Chyavanprash Forte Tab	A	68000	4.4
Desa-Ambubidhi Syng	A	65000	5.5
Calbert K27 Cap	A	60000	6.6
Rabizine DS8 Cap	A	58000	7.7
Itra-Aloset 200mg Ca	A	55000	8.8
Alonita FX Tab	A	49000	9.9
Calbert D3 60R Cap	A	48000	11.0
Alonita L Tab (100s)	A	47000	12.1
Kecurus Shampoo 100ml	A	47000	13.2
Rabulitem DS8 Cap	A	45000	14.3
Mochert CT 625 Tab	A	44000	15.4
Chyavanprash 500g	A	44000	16.5
Skinshine Ointment .1	B	37000	17.6
Lycoveen Cap	B	37000	18.7
Scillex CX Tab	B	34000	19.8
Panderr D Tab	B	33000	20.9
Dolo Plus Syng	B	32000	22.0
Intapopan Syng 600m	B	25000	23.1
Patohort 40 Tab	B	25000	24.2
Alonita Plus Cap	C	22000	25.3
Mores Eye Drop	C	18000	26.4
Montmor LC Syng 600m	C	18000	27.5
Clebsan GM 10g Cream	C	18000	28.6
Mosch D Eye Drop	C	16000	29.7
Malaise Tab	C	15000	30.8
Decidite Ju Juul	C	15000	31.9
Gentlele 100ml Ju	C	7000	33.0

5.2 FSN Inventory Movement & Expiry Risk Findings

Figure 4: FSN (Fast-Slow-Non Moving) Inventory Classification
Anand Pharma (Jan-June 2026)



ABC-FSN Matrix: Category A - Slow Moving items (10 SKUs) form the largest segment, requiring strict reorder point management to prevent stockouts.

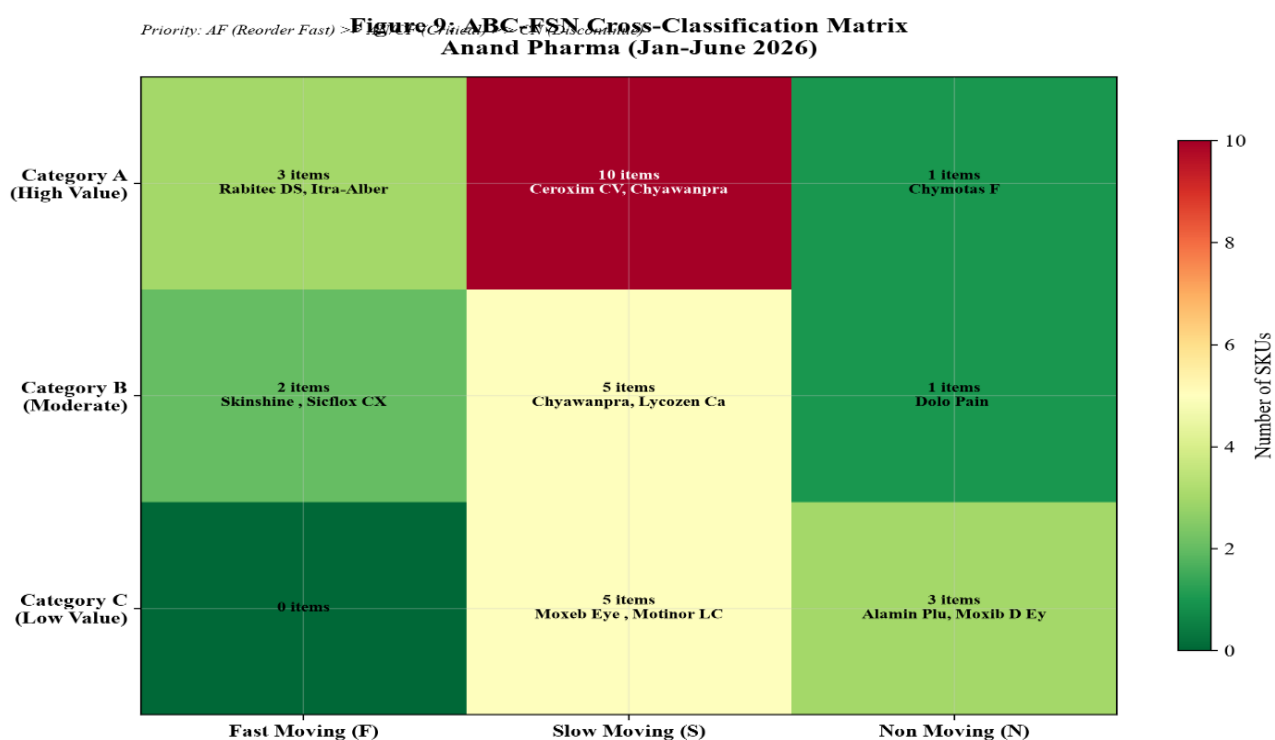


Figure 4: ABC-FSN Priority Matrix

5.3 Monthly Sales Trends & Category Distribution

Monthly Store Revenue: Jan (₹2,20,925), Feb (₹2,14,824), Mar (₹2,07,518), Apr (₹1,97,038 - lowest), May (₹2,43,998 - 23.8% pre-monsoon surge), Jun (₹2,21,352). Antibiotics (₹2,87,988) and Gastric Care (₹1,63,259) drive >34% of revenue.

Figure 1: Monthly Revenue Trend & Units Sold (Jan-June 2026)
Anand Pharma, Darbhanga

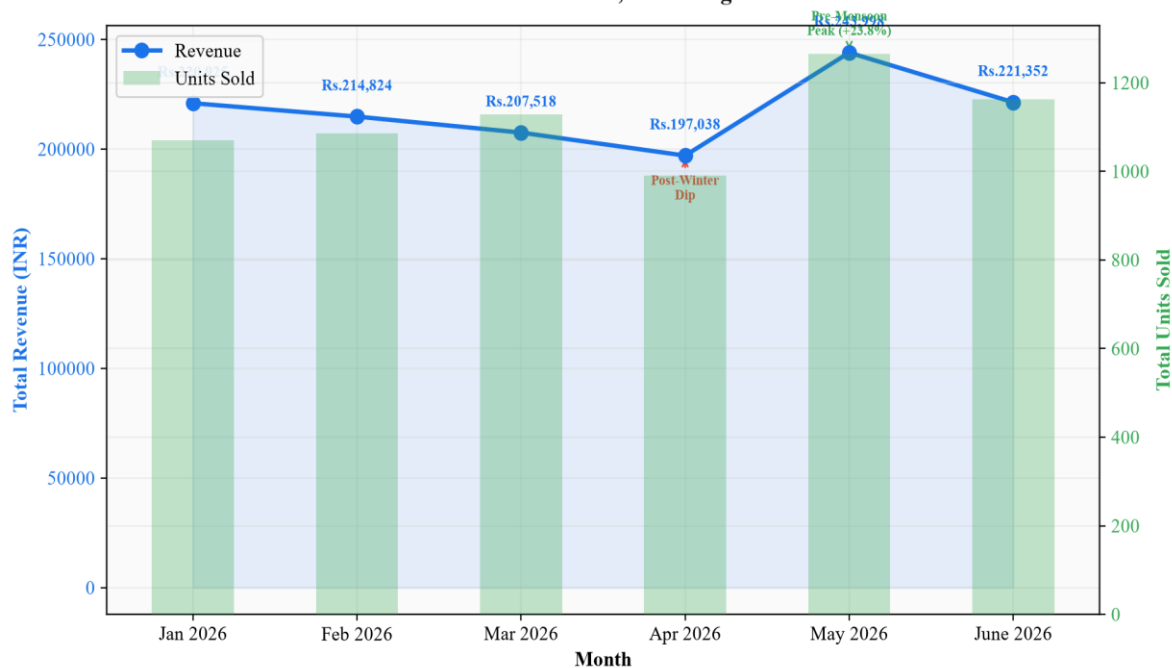


Figure 5: Monthly Revenue & Sales Volume Trend

Figure 6: Revenue Distribution by Therapeutic Category
Anand Pharma (Jan-June 2026)

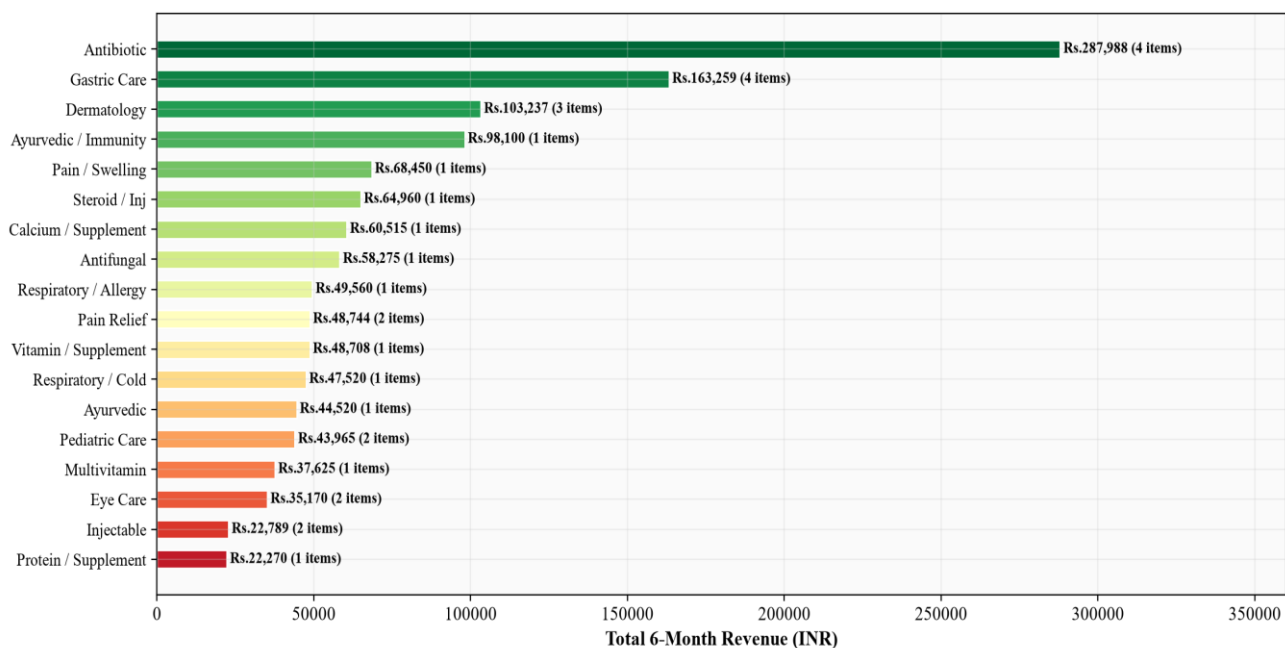


Figure 6: Revenue by Therapeutic Category

5.4 Moving Average Demand Forecasting Results

July 2026 Demand Forecasts (3-Month SMA): Ceroxim CV 500 (45 strips), Abzolid 600 (37 strips), Chyawanprash 2KG (32 units - seasonal drop), Chymotas Forte (26 strips). SMA provides a conservative, lag-adjusted baseline for advance ordering.

Figure 5: 3-Month Moving Average Demand Forecast (Top 6 Category A Medicines)
Anand Pharma (Jan-June 2026)

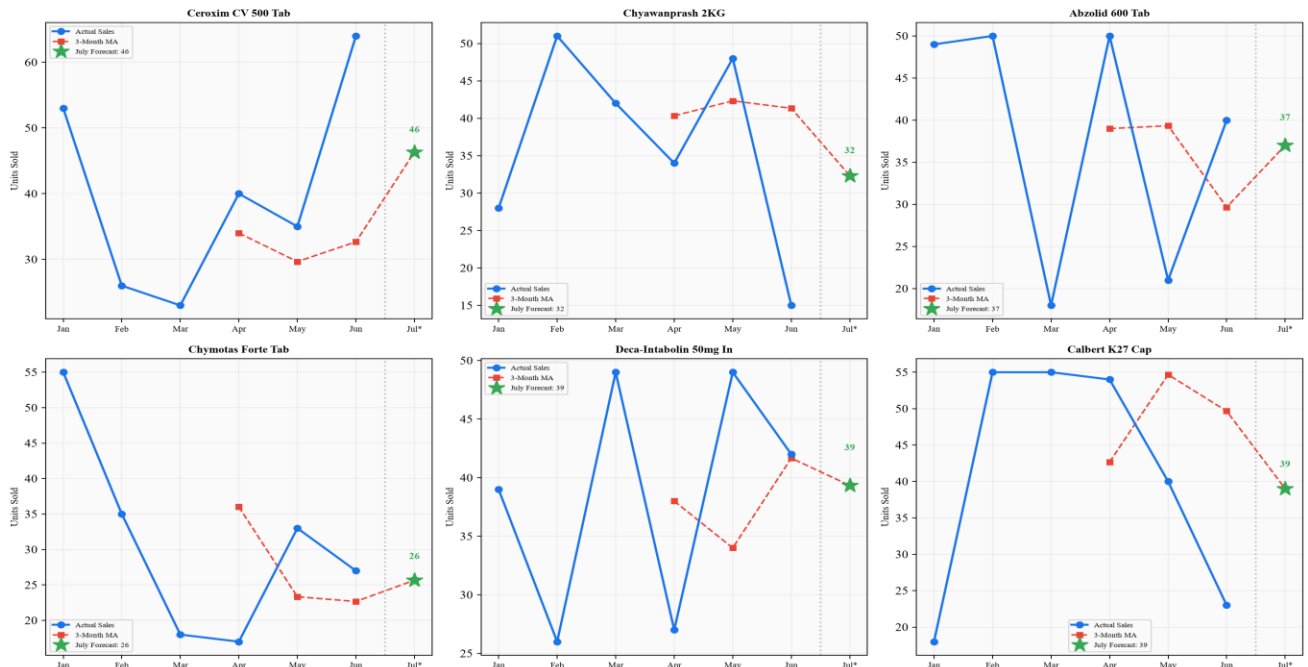


Figure 7: 3-Month Moving Average Forecast (Top SKUs)

Figure 7: Sales Quantity vs Revenue (Bubble Size = MRP)
Anand Pharma (Jan-June 2026)

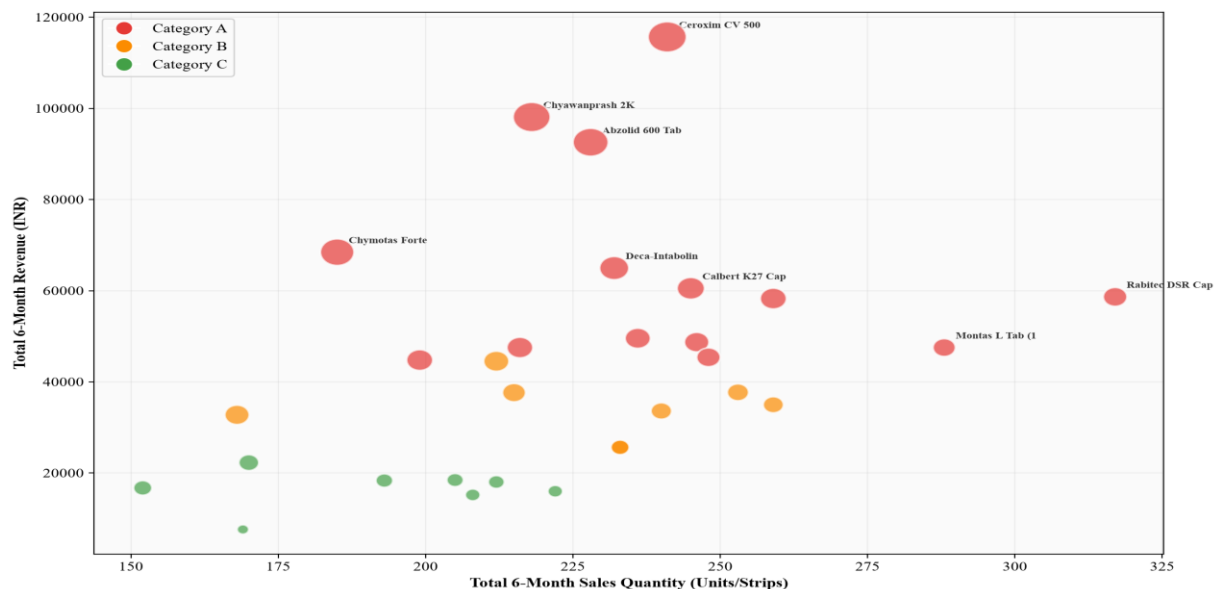


Figure 8: Quantity vs Revenue Bubble Chart (Size = MRP)

5.5 Conclusion

1. Category A Safety Buffer Stocking (Addresses PS1): Maintain a 15-day safety buffer for Category A SKUs (Ceroxim CV, Abzolid) using $\text{Reorder Point} = \text{Avg Daily Sales} \times \text{Lead Time} + \text{Buffer}$.
2. Pre-Expiry Credit Returns (Addresses PS2): Implement a 60-day pre-expiry audit to return 5 non-moving SKUs (Gentalab Inj, Dexalife Inj) to distributor M/s. Sonu Pharma for credit notes.
3. Seasonal Advance Ordering (Addresses PS3): Increase Antibiotic and Gastric orders by 15-20% in April to capture the May-June pre-monsoon demand surge.
4. Digital Stock Audit Transition: Transition from manual registers to weekly Excel digital tracking for automated low-stock alerts.

Figure 8: Profit Margin Analysis (MRP vs Purchase Price)
Anand Pharma - 30 Medicine SKUs

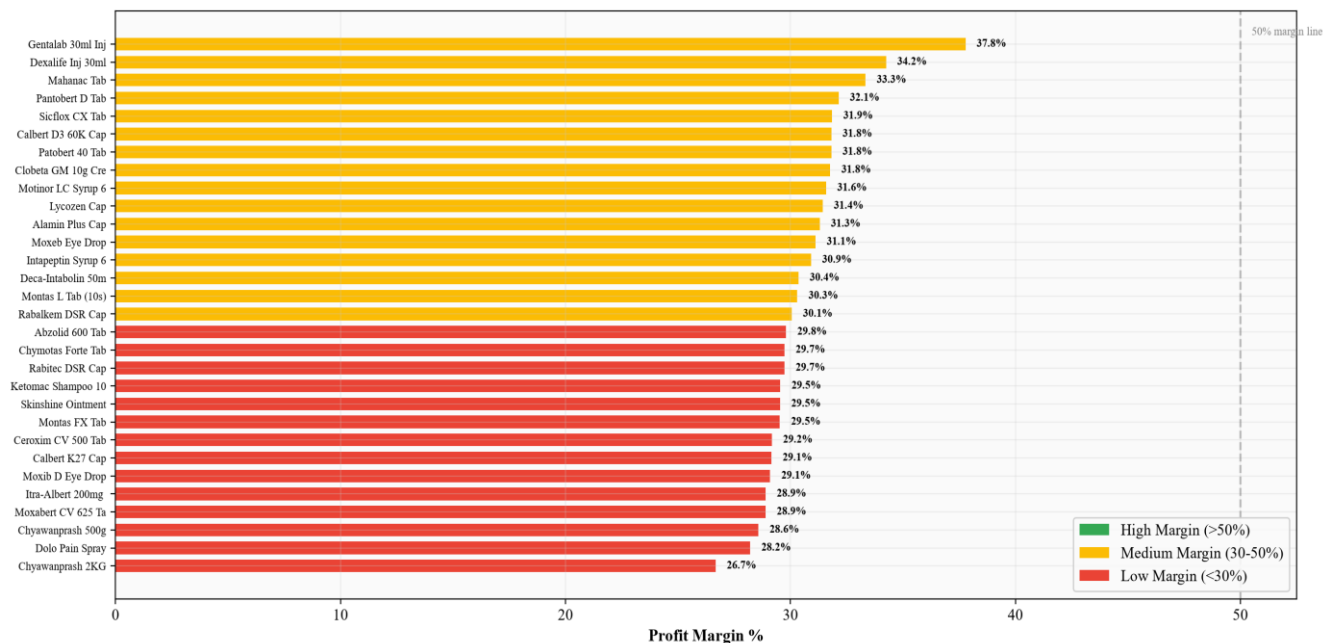


Figure 9: Profit Margin Analysis across 30 SKUs

Figure 10: Stock Movement Pattern (Top 5 Category A Medicines)
Opening Stock, Purchase, Sales & Closing Stock

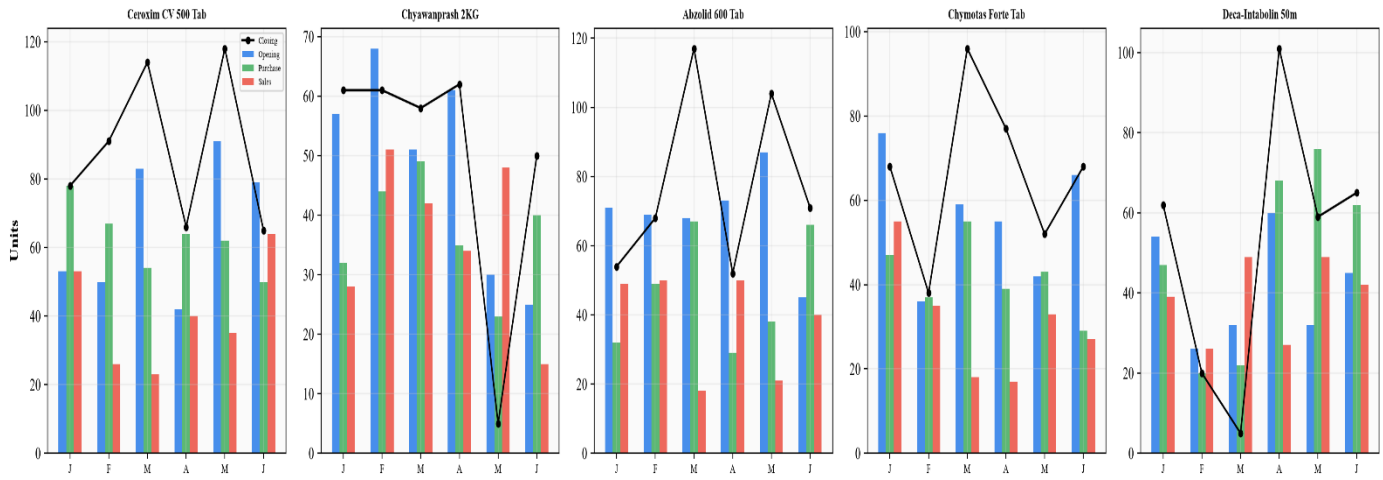


Figure 10: Stock Movement Pattern (Top 5 SKUs)